II. The coarse category. Coarse Cohomology

2.0. Reminder:

- $(G,\cdot)$ f.g. group $\longleftrightarrow$ (s. fin. gen. set) $\Cay(G,S)$ connected graph, (G,d_S) metric space

Algebra ..... Geometry at infinity

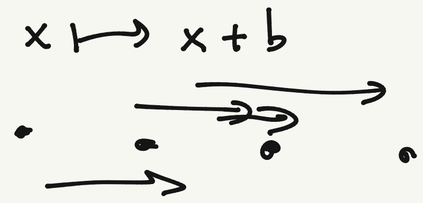
- Quasi-isometries: $X \overset{QI}{\sim} Y$
  if X, Y look the same from far away.
- From far away, they way (G,d_S) looks does not depend on S:

$$\left[S, S' \text{ fin. gen. sets} \Rightarrow id_G : (G,d_{S'}) \xrightarrow{QI} (G,d_S). \right]$$

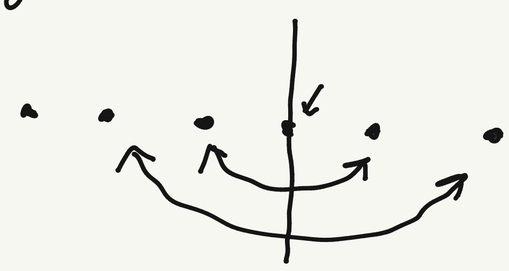
- Pending exercise: Recognize $\mathrm{Isom}(\mathbb{Z}) = D_\infty$ from far away.

Two kinds of isom. order-preserving; translations

$x \longmapsto x+b$



order-reversing: symmetries

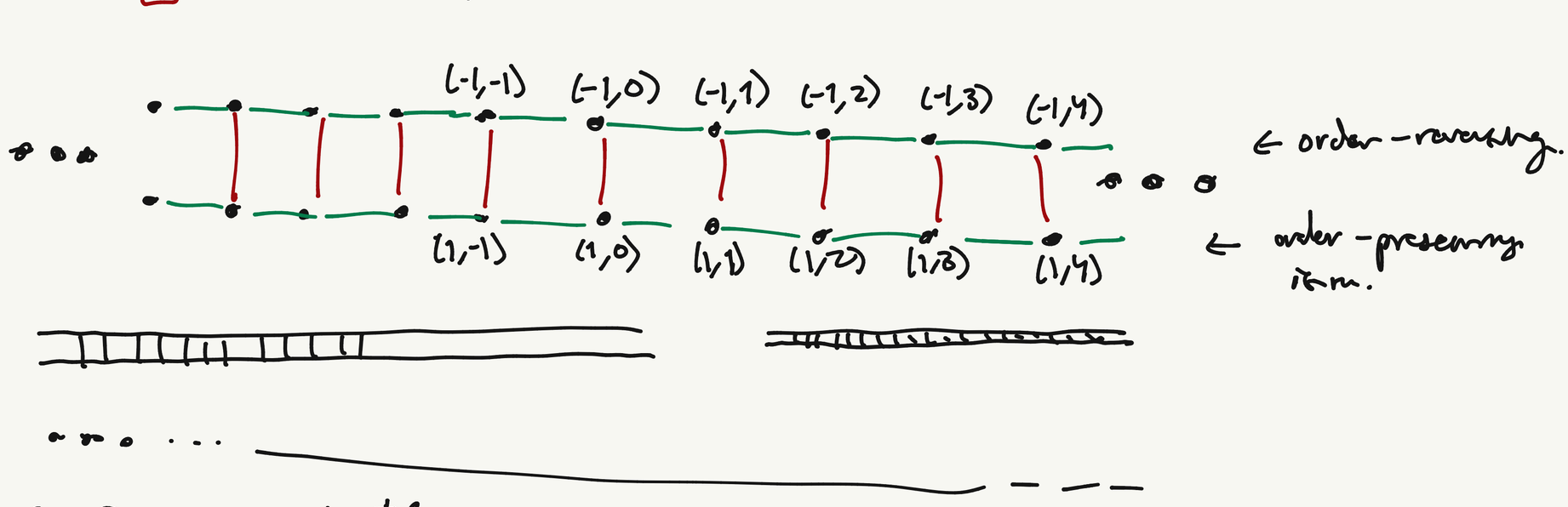


$\mathrm{Isom}(\mathbb{Z}) =$

$= D_\infty = \langle t, s \rangle$

$t = (x \mapsto x+1)$

$s = (x \mapsto -x)$



$G = D_\infty$, $S = \{t^{\pm 1}, s\}$ i $\Cay(G,S) \overset{QI}{\sim} \mathbb{R} \overset{QI}{\sim} \mathbb{Z}$.

$$\mathrm{Isom}(\mathbb{Z}) = \{ f_{a,b} : \mathbb{Z} \to \mathbb{Z},\ x \mapsto ax+b \mid a \in \{\pm 1\},\ b \in \mathbb{Z} \}$$

$f_{a,b} \sim (a,b)$. $\qquad f_{1,0} : x \mapsto x$

$$f_{a',b'} \circ f_{a,b}(x) = a'(ax+b) + b',$$

$$(a',b')(a,b) = (a'a,\ a'b+b').$$

2.1. Schwarz - Milnor lemma

- Let G be a group acting by isometries on a proper geodesic metric space (X,d). Assume $G \curvearrowright X$ is:
  -) proper: $\forall K \subseteq X$ compact, $\{g \in G \mid gK \cap K \neq \emptyset\}$ is finite.
  -) cocompact: X/G is compact.

  Then G is f.g. and $G \underset{QI}{\sim} X$. Moreover, for every $x_0 \in X$,

$$f : G \longrightarrow X$$
$$g \longmapsto g \cdot x_0$$

  is a quasi-isometry.

- (M,g) closed connected Riem. manifold.

  Let $p : \tilde{M} \to M$ local isometry. $\tilde{M}$ is a Riem. structure

  $\pi_1(M) \curvearrowright \tilde{M}$ satisfies the hypothesis above. Then

  -) $\pi_1(M)$ is finitely generated
  -) $\pi_1(M) \underset{QI}{\sim} \tilde{M}$.

- $M = \mathbb{S}^1$ closed Riem. mnfld.

$p : \mathbb{R} \to \mathbb{S}^1$; By Schwarz – Milnor,

$$\underbrace{\pi_1(\mathbb{S}^1)}_{\mathbb{Z}} \underset{QI}{\sim} \mathbb{R}$$



- M is a closed surface, orientable of genus $g \geq 2$.

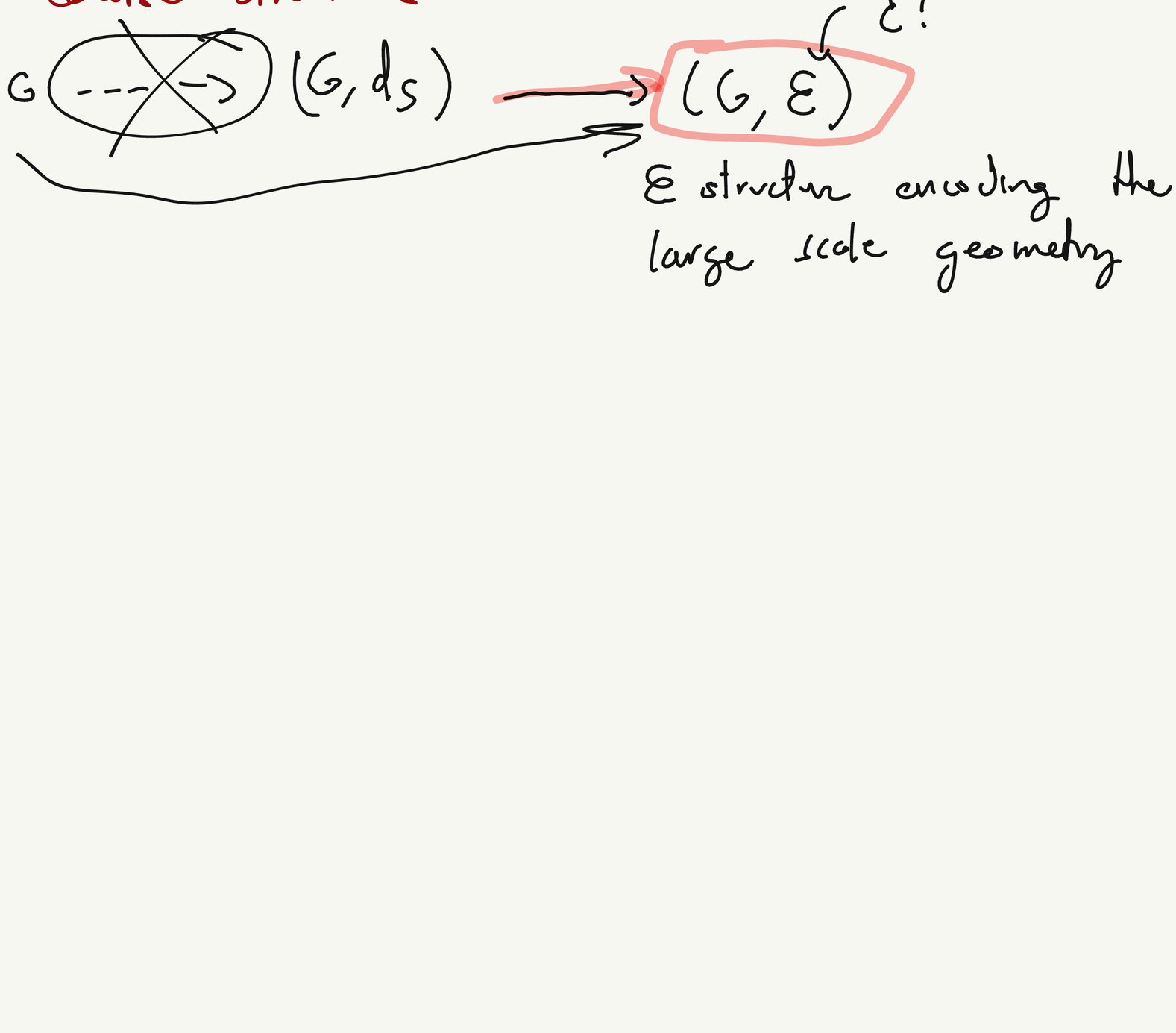
$M =$ 

$\widetilde{M} = \mathbb{H}^2$

$$\pi_1(M) \underset{QI}{\sim} \mathbb{H}^2.$$

- $M = \mathbb{S}^1 \times \mathbb{S}^1$; $\widetilde{M} = \mathbb{R}^2$; $\underbrace{\pi_1(\mathbb{S}^1 \times \mathbb{S}^1)}_{\mathbb{Z} \times \mathbb{Z}} \underset{QI}{\sim} \mathbb{R}^2$.

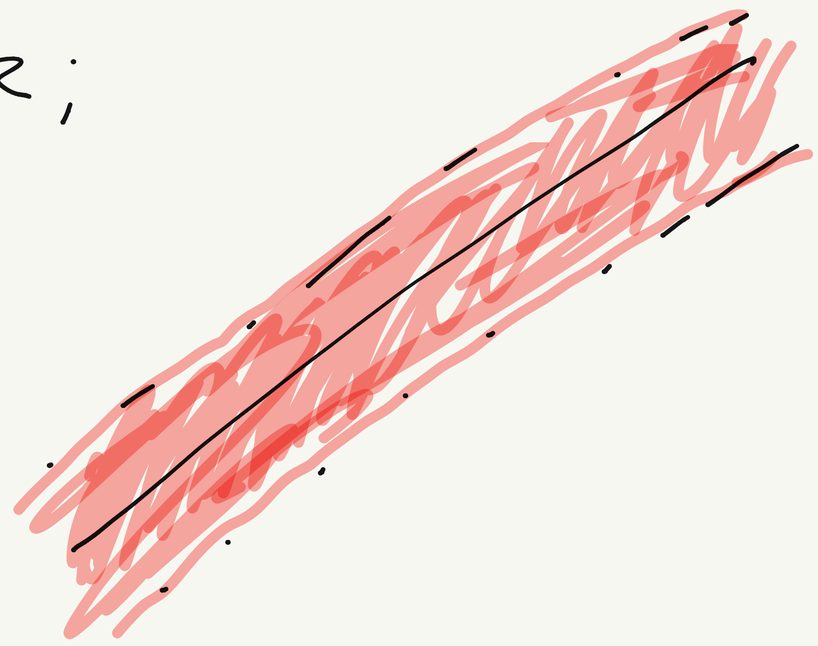
2.2. Coarse structures.

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• (X,d) metric space. For each $r>0$,

$$T_r := \{(x_1,x_2) \in X\times X \mid d(x_1,x_2) \le r\}$$

If $X=\mathbb{R}$;



i) $s \le r$, $T_s \subseteq T_r$.

ii) $r_1, r_2 > 0$, $T_{r_1} \cup T_{r_2} \overset{(=)}{\subseteq} T_{\max\{r_1,r_2\}}$

iii) $\Delta \subseteq T_r \quad \forall r>0$.

iv) (Triangle inequality.)

Given $x_1, x_2, x_3 \in X$,

$T_{r_1} ; T_{r_2} \subseteq T_{r_1+r_2}$?

$(x_1,x_2) ; (x_2,x_3) = (x_1,x_3)$

$x_1 \to x_2 \longrightarrow x_3$

$$T_{r_1} ; T_{r_2} = \{(x_1,x_3) \mid \exists x_2 \in X \text{ s.t. } d(x_1,x_2)\le r_1,\ d(x_2,x_3)\le r_2\}$$

$d(x_1,x_3) \le r_1 + r_2$.

v) If $(x_1,x_2) \in X\times X$, $(x_1,x_2)^{-1} = (x_2,x_1)$.

If $E \subseteq X\times X$, $E^{-1} = \{(x_2,x_1) \mid (x_1,x_2)\in E\}$.

$T_r = T_r^{-1}$.

- **Defn**: Let X be a set. A <u>coarse structure</u> on X is a family $\mathcal{E}$ of subsets of $X \times X$ ("entourages" or "controlled sets") satisfying:

  i) $E_1 \subseteq E_2 \in \mathcal{E} \Rightarrow E_1 \in \mathcal{E}$.

  ii) $E_1, E_2 \in \mathcal{E} \Rightarrow E_1 \cup E_2 \in \mathcal{E}$.

  iii) $\Delta_X = \{(x,x) \mid x \in X\} \in \mathcal{E}$.

  iv) $E_1, E_2 \in \mathcal{E}$,

  $\Rightarrow E_1 ; E_2 = \{(x_1, x_3) \mid \exists x_2 \in X \text{ s.t. } (x_1,x_2) \in E_1, (x_2,x_3) \in E_2\} \in \mathcal{E}$

  $x_1 \to x_2 \to x_3$

  v) $E \in \mathcal{E} \Rightarrow E^{-1} \in \mathcal{E}$.

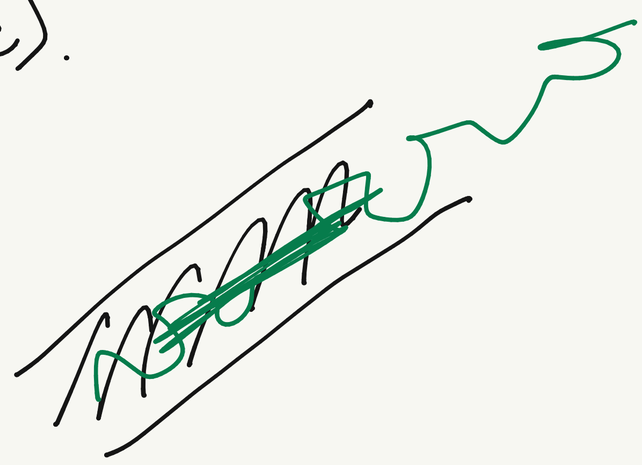
A <u>coarse space</u> is a pair $(X, \mathcal{E})$.

2.3. Examples

- (Bded) metric coarse structure:
  Let (X,d) be a metric space. Then

$$\mathcal{E}_d := \{E \subseteq X \times X \mid \exists r > 0 \text{ s.t. } \underbrace{E \subseteq T_r}_{d(x_1,x_2) \le r \ \forall (x_1,x_2) \in E}\}$$

$$= \{E \subseteq X \times X \mid \sup_{(x_1,x_2) \in E} d(x_1,x_2) < \infty\}$$



$(G, \cdot)$ --- > (G, d_S) ⟶ $(G, \mathcal{E})$

f.g. — metric space — Bded metric structure — coarse space

- C_0 - metric coarse structure.

  (X,d) metric space. Let $E \subseteq X^2$. Then,

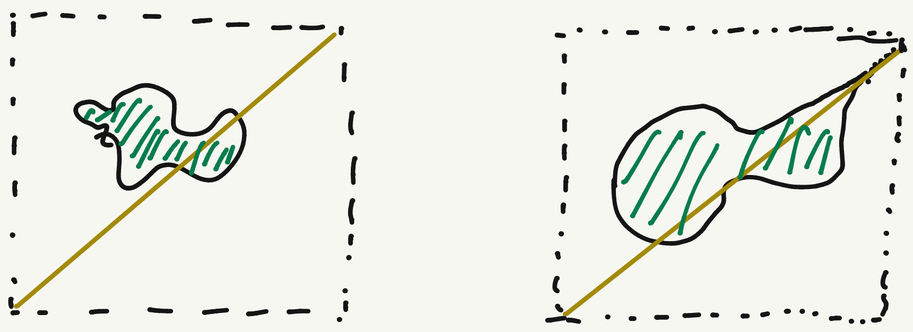
  $E \in \mathcal{E}_{C_0} \iff d|_E : E \to \mathbb{R}$ "tends to zero at infinity"

  $\iff \forall \varepsilon > 0 \quad \exists K \subseteq X$ cpt such that

  $d|_{(E \setminus K \times K)} < \varepsilon$.

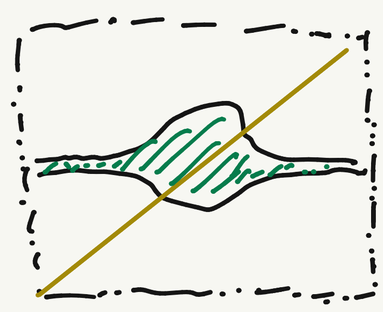
  $\big((x_1,x_2) \in E$ and $(x_1 \notin K$ or $x_2 \notin K)\big) \Rightarrow d(x_1,x_2) < \varepsilon$.

  $(X,d) = (\mathbb{R}, d_{\text{stdd}})$ $\quad \mathbb{R} \overset{\sim}{\subset} (0,1)$

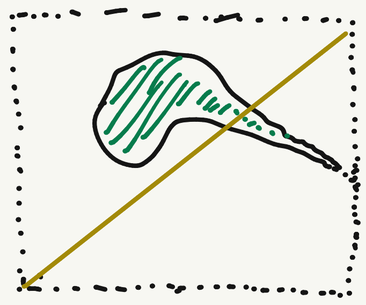


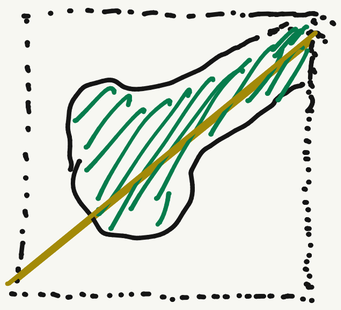
Controlled $\quad E \subseteq X^2$

$\Delta = \{(x,x) \mid x \in \mathbb{R}\}$



Not controlled





- Topological coarse structures associated to compactifications.

  Let $X \in \mathrm{Top}$ paracompact, loc. cpt. Hausdorff

  ((X,d) proper metric spaces satisfy this).

  A compactification is:

  $i : X \hookrightarrow \overline{X}$ cpt. Hausdorff

  i: top. emb.

  $i(X)$ open dense in $\overline{X}$.

  $X \subseteq \overline{X}$. The corona of the compactification is

  $\partial X = \overline{X} \setminus X$ (points at infinity)

$$\mathcal{E}_i = \{ \text{"continuously controlled sets by } i\text{"} \} =$$

$$= \{ E \subseteq X^2 \mid \overline{E} \setminus (X \times X) \subseteq \Delta_{\partial X} \} =$$

in $\overline{X} \times \overline{X}$

$$= \{ E \subseteq X^2 \mid \overline{E} \cap (\partial X \times X \cup X \times \partial X) \subseteq \Delta_{\partial X} \} =$$

$$= \{ E \subseteq X^2 \mid ((w_1, w_2) \in \overline{E} \text{ and either } w_1 \in \partial X \text{ or } w_2 \in \partial X) \Rightarrow w_1 = w_2 \}$$

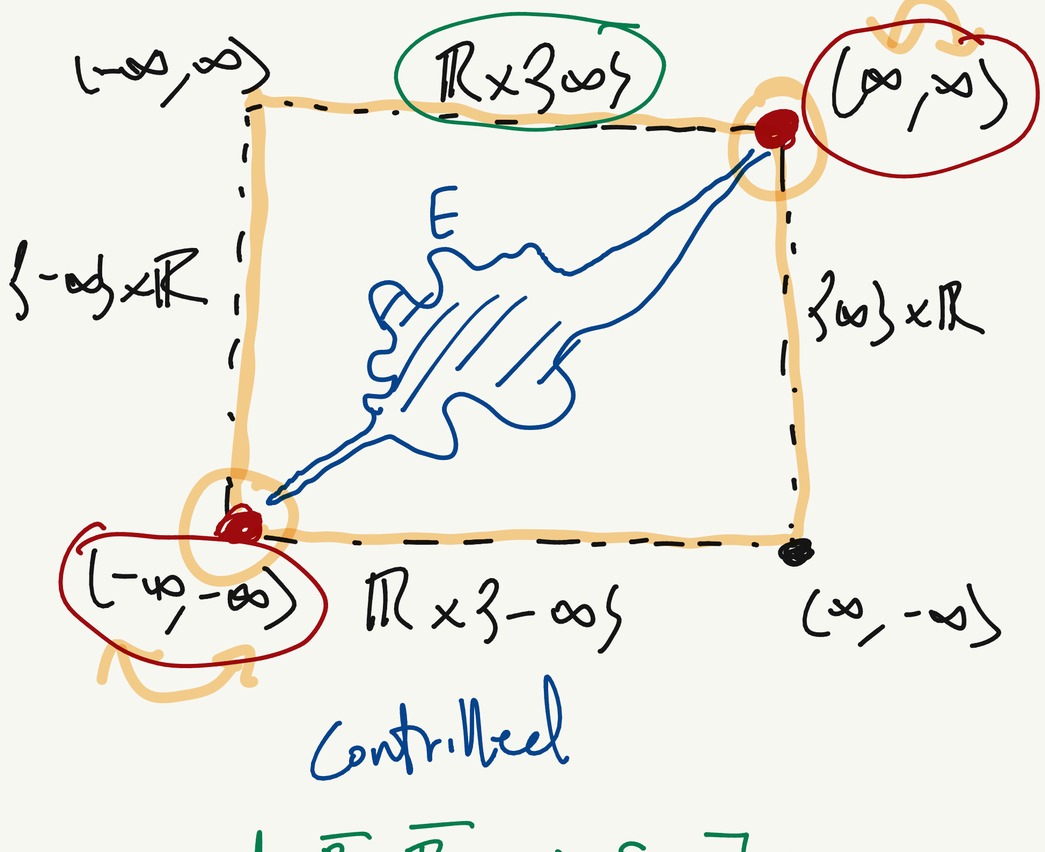
$\Rightarrow \mathcal{E} \subseteq \mathcal{P}(X \times X)$

$\mathcal{E} = \mathcal{P}(X \times X)$
X bded $X \sim *$

Example: $X = \mathbb{R}$, two compactifications:

$\hookrightarrow i : \mathbb{R} \hookrightarrow \overline{\mathbb{R}} = [-\infty, \infty]$ identified w/ $(0,1) \hookrightarrow [0,1]$

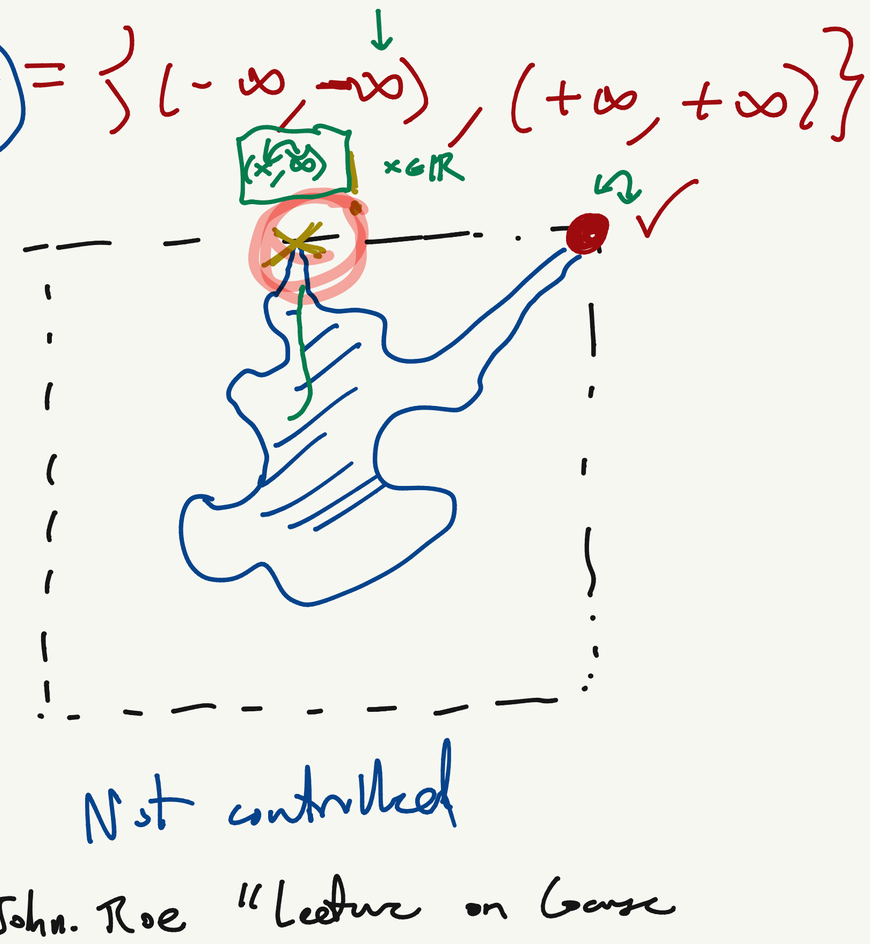
$E \subseteq \mathbb{R}^2 \subseteq \overline{\mathbb{R}}^2$



Controlled

$d : \overline{\mathbb{R}} \times \overline{\mathbb{R}} \to [0, \infty]$ /

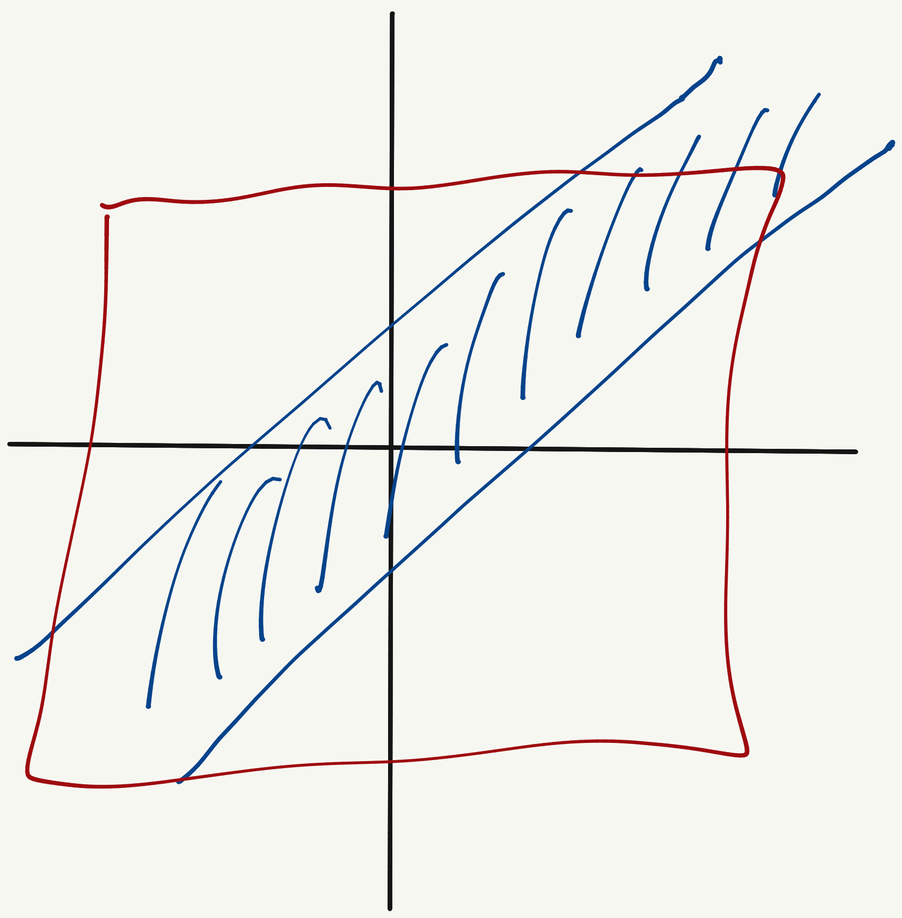
$\Delta_{\partial X} = \{(-\infty, -\infty), (+\infty, +\infty)\}$



Not controlled

John. Roe "Lecture on Coarse Geometry" Ch. 2.

$\mathbb{R} \cong (0,1)$ $\qquad E \subseteq \mathbb{R}^2$



G finite, $G \curvearrowright S$ $\qquad G \underset{QI}{\sim} *$ $\qquad$ (Schwarz – Milnor)